DB_Project_Assignment_9- ERD to Relational Model, and Normalization

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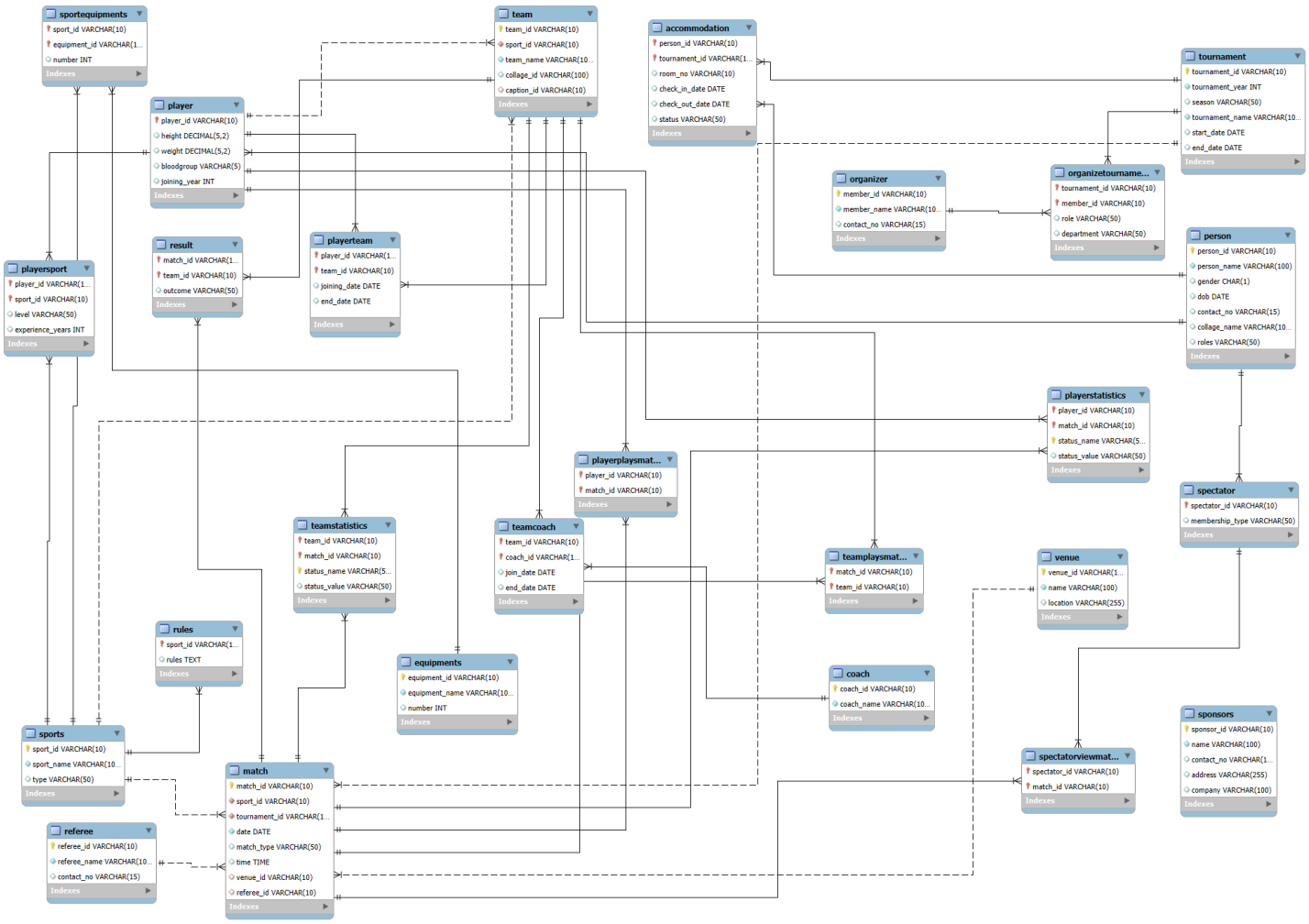
Relational Diagram



1. Entity Sets (Strong & Weak)

1. **Person** (person_id, person_name, gender, dob, contact_no, collage_name, roles)
2. **Tournament** (tournament_id, tournament_year, season, tournament_name, start_date, end_date)
3. **Sponsors** (sponsor_id, name, contact_no, address, company)
4. **Sports** (sport_id, sport_name, type)
5. **Equipments** (equipment_id, equipment_name, number)
6. **Coach** (coach_id, coach_name)
7. **Venue** (venue_id, name, location)
8. **Referee** (referee_id, referee_name, contact_no)
9. **Organizer** (member_id, member_name, contact_no)
10. **Team** (team_id, sport_id(FK), team_name, collage_name, caption_id)

    FKs: sport_id references Sports(sport_id), caption_id references Player(player_id)

11. **Match** (match_id, sport_id(FK), tournament_id(FK), date, match_type, time, venue_id(FK), referee_id(FK))
12. **Rules** (sport_id, rules)
13. **Accommodation** (person_id(FK),tournament_id(FK), room_no, check_in_date, check_out_date, status)

2. Entity Sets (Subclasses/Specialization)

These tables inherit the primary key from the **Person** superclass.

14. **Player** (player_id (FK), height, weight, bloodgroup, joining_year)

FK: player_id references Person(person_id) (This is a 1:1 relationship, so the PK is also the FK).

15. **Spectator** (spectator_id, membership_type)

FK: spectator_id references Person(person_id) (PK/FK).

3. Relationship Sets (Junction Tables)

16.**OrganizeTournament** (tournament_id,member_id, role, department)

17. **SpectatorViewMatch** (spectator_id,match_id)

FKs: spectator_id, match_id

18**. PlayerPlaysMatch** (player_id,match_id)

- *FKs: player_id, match_id*

19. **SportEquipments** (sport_id,equipment_id, number)

- *FKs: sport_id, equipment_id*

20.**PlayerSport** (player_id,sport_id, level, experience_years)

- *FKs: player_id, sport_id*

21. **PlayerTeam** (player_id,team_id, joining_date, end_date)

- *FKs: player_id, team_id*

22. **TeamPlaysMatch** (match_id,team_id)

- *FKs: match_id, team_id*

23. **TeamCoach** (team_id,coach_id, join_date, end_date)

- *FKs: team_id, coach_id*

24. **PlayerStatistics** (player_id,match_id,status_name, status_value)

- *FKs: player_id, match_id*

25. **TeamStatistics** (team_id,match_id,status_name, status_value)

- *FKs: team_id, match_id*

26. **Result** (match_id,team_id, outcome)

- *FKs: match_id, team_id*

27. **SponsorsTournament** (tournament_id(FK) , sponsor_id(FK) , budget)

List of all dependencies:

PK-primary Key
FK-foreign key
FD-Function Dependencies

1. Person

Schema:
`Person(person_id, person_name, gender, dob, contact_no, collage_name, roles)`

PK: person_id

But contact_no can also be another candidate key for a primary key. Here, we get a one-to-one functional dependency that is not redundant.
FK: none
FDs: person_id → person_name, gender, dob, contact_no, collage_name, roles
(Each person_id uniquely identifies one person.)

[Each person can register through only one contact_no, so we do not get multi-valued dependency.]

2. Tournament

Schema:

`Tournament(tournament_id, season, tournament_name, start_date, end_date)`

PK: tournament_id

{**CD:** start_date, because start_date is a candidate key and uniquely identifies each tuple in the Tournament.}

FDs:tournament_id → tournament_year, season, tournament_name, start_date, end_date

3. Sponsors

Schema:
`Sponsors(sponsor_id, name, contact_no, address, company)`

PK: sponsor_id
FDs: sponsor_id → name, contact_no, address, company
TD: The company defines the address.

4. Sports

Schema:
`Sports(sport_id, sport_name, type)`

PK: sport_id
FDs:sport_id → sport_name, type

CD: sport_name is a candidate key. It implies CK dependency.

5. EquipmentSchema:

`Equipments(equipment_id, equipment_name, number)`

PK: equipment_id
FDs:equipment_id → equipment_name, number

[equipment_name as well as equipment_id can act as primary key, so we can remove equipment_id further during Normalization.}

6. Coach

Schema:
`Coach(coach_id, coach_name)`

PK: coach_id
FDs:coach_id → coach_name

7. Venue

Schema:
`Venue(venue_id, name, location)`

PK: venue_id
FDs:venue_id → name, location

8. Referee

Schema:
`Referee(referee_id, referee_name, contact_no)`

`{contact_no is unique, avoiding multi-valued dependency.}`

PK: referee_id
FDs:referee_id → referee_name, contact_no

9. Organizer

Schema:
`Organizer(member_id, member_name, contact_no)`

PK: member_id
FDs:member_id → member_name, contact_no

10. Team

Schema:
`Team(team_id, sport_id(FK), team_name,collage_name,captain_id(FK))`

PK: team_id
FKs: sport_id → Sports(sport_id)
captain_id → Player(player_id)

FDs:team_id → sport_id, team_name, collage_id, caption_id

11. Match

Schema:
`Match(match_id, sport_id(FK), tournament_id(FK), date, match_type, time, venue_id(FK), referee_id(FK))`

PK: match_id
FKs: sport_id → Sports(sport_id)
tournament_id → Tournament(tournament_id)

venue_id → Venue(venue_id)
referee_id → Referee(referee_id)

FDs:match_id → sport_id, tournament_id, date, match_type, time, venue_id, referee_id

12. Rule:

Schema:
`Rules(sport_id(FK), rules)`

PK: sport_id
FK: sport_id → Sports(sport_id)

FDs: sport_id → rules

13. Accommodation

Schema:
`Accommodation(person_id(FK), tournament_id(FK), room_no, check_in_date, check_out_date, status)`

PK: (person_id, tournament_id)
FKs: person_id → Person(person_id)
tournament_id → Tournament(tournament_id)
FDs:(person_id, tournament_id) → room_no, check_in_date, check_out_date, status

14. Player

Schema:
`Player(player_id(FK), height, weight, bloodgroup, joining_year)`

PK: player_id
FK: player_id → Person(person_id)

FDs:player_id → height, weight, bloodgroup, joining_year

15. Spectator

Schema:
`Spectator(spectator_id(FK), membership_type)`

PK: spectator_id
FK: spectator_id → Person(person_id)

FDs:spectator_id → membership_type

16. Organize Tournament

Schema:
`OrganizeTournament(tournament_id(FK), member_id(FK), role, department)`

PK: (tournament_id, member_id)
FKs:tournament_id → Tournament(tournament_id)
member_id → Organizer(member_id)
FDs:(tournament_id, member_id) → role, department

17. SpectatorViewMatch

Schema:
`SpectatorViewMatch(spectator_id(FK), match_id(FK))`

PK: (spectator_id, match_id)
FKs: spectator_id → Spectator(spectator_id)
match_id → Match(match_id)

FDs:(spectator_id, match_id) → (no additional attributes)

18. PlayerPlaysMatch

Schema:
`PlayerPlaysMatch(player_id(FK), match_id(FK))`

PK: (player_id, match_id)
FKs:player_id → Player(player_id)
match_id → Match(match_id)

FDs:(player_id, match_id) → (no additional attributes)

19. Sport Equipment

Schema:
`SportEquipments(sport_id(FK), equipment_id(FK), number)`

PK: (sport_id, equipment_id)
FKs: sport_id → Sports(sport_id)
equipment_id → Equipments(equipment_id)

FDs: (sport_id, equipment_id) → number

20. PlayerSport

Schema:
`PlayerSport(player_id(FK), sport_id(FK), level, experience_years)`

PK: (player_id, sport_id)
FKs: player_id → Player(player_id)
sport_id → Sports(sport_id)
FDs:(player_id, sport_id) → level, experience_years

21. PlayerTeam

Schema:
`PlayerTeam(player_id(FK), team_id(FK), joining_date, end_date)`

PK: (player_id, team_id)
FKs: player_id → Player(player_id)
team_id → Team(team_id)
FDs: (player_id, team_id) → joining_date, end_date

22. TeamPlaysMatch

Schema:
`TeamPlaysMatch(match_id(FK), team_id(FK))`

PK: (match_id, team_id)
FKs: match_id → Match(match_id)
team_id → Team(team_id)

FDs: (match_id, team_id) → (no additional attributes)

23. TeamCoach

Schema:
`TeamCoach(team_id(FK), coach_id(FK), join_date, end_date)`

PK: (team_id, coach_id)
FKs: team_id → Team(team_id)
coach_id → Coach(coach_id)

FDs: (team_id, coach_id) → join_date, end_date

24. PlayerStatistics

Schema:
`PlayerStatistics(player_id(FK), match_id(FK), status_name, status_value)`

PK: (player_id, match_id, status_name)
FKs:player_id → Player(player_id)
match_id → Match(match_id)

FDs:(player_id, match_id, status_name) → status_value

25. TeamStatistics

Schema:
`TeamStatistics(team_id(FK), match_id(FK), status_name, status_value)`

PK: (team_id, match_id, status_name)
FKs: team_id → Team(team_id)
match_id → Match(match_id)

FDs: (team_id, match_id, status_name) → status_value

26. Result

Schema:
`Result(match_id(FK), team_id(FK), outcome)`

PK: (match_id, team_id)
FKs: match_id → Match(match_id)
team_id → Team(team_id)

FDs:(match_id, team_id) → outcome

No.	Relation Name	PK	Partial Dependencies	Transitive Dependencies
1	**Person**(person_id, person_name, gender, dob, contact_no, collage_name, roles)	person_id	None (simple key)	None
2	**Tournament**(tournament_id, tournament_year, season, tournament_name, start_date, end_date)	tournament_id	None	None
3	**Sponsors**(sponsor_id, name, contact_no, address, company)	sponsor_id	None	None
4	**Sports**(sport_id, sport_name, type)	sport_id	None	None
5	**Equipments**(equipment_id, equipment_name, number)	equipment_id	None	None
6	**Coach**(coach_id, coach_name)	coach_id	None	None
7	**Venue**(venue_id, name, location)	venue_id	None	None
8	**Referee**(referee_id, referee_name, contact_no)	referee_id	None	None
9	**Organizer**(member_id, member_name, contact_no)	member_id	None	None

10	**Team**(team_id, sport_id, team_name, collage_id, caption_id)	team_id	None	None
11	**Match**(match_id, sport_id, tournament_id, date, match_type, time, venue_id, referee_id)	match_id	None	None
12	**Rules**(sport_id, rules)	sport_id	None	None
13	**Accommodation**(person_id, tournament_id, room_no, check_in_date, check_out_date, status)	(person_id, tournament_id)	None (attributes depend on both keys)	None
14	**Player**(player_id, height, weight, bloodgroup, joining_year)	player_id	None	None
15	**Spectator**(spectator_id, membership_type)	spectator_id	None	None
16	**OrganizeTournament**(tournament_id, member_id, role, department)	(tournament_id, member_id)	None	None
17	**SpectatorViewMatch**(spectator_id, match_id)	(spectator_id, match_id)	None	None
18	**PlayerPlaysMatch**(player_id, match_id)	(player_id, match_id)	None	None
19	**SportEquipments**(sport_id, equipment_id, number)	(sport_id, equipment_id)	None	None
20	**PlayerSport**(player_id, sport_id, level, experience_years)	(player_id, sport_id)	None	None
21	**PlayerTeam**(player_id, team_id, joining_date, end_date)	(player_id, team_id)	None	None
22	**TeamPlaysMatch**(match_id, team_id)	(match_id, team_id)	None	None

23	**TeamCoach**(team_id, coach_id, join_date, end_date)	(team_id, coach_id)	None	None
24	**PlayerStatistics**(player_id, match_id, status_name, status_value)	(player_id, match_id, status_name)	None	None
25	**TeamStatistics**(team_id, match_id, status_name, status_value)	(team_id, match_id, status_name)	None	None
26	**Result**(match_id, team_id, outcome)	(match_id, team_id)	None	None

NORMALIZATION :

As we can see, there are no other dependencies other than Primary Key Dependency, so the above schemas listed in the table are already in 1NF, 2NF, and 3NF form.

BCNF:

For BCNF:

1. Sponsors: (sponsor_id, name, contact_no, address, company)
   It violates BCNF as company->address
   Solution: It is split into
   a)Sponsors(sponsor_id,name,contact_no,company)
   b)Company(company,address)

2. Sports
   sport_id->sport_name,type and sport_name->type
   So, sport_name is not a super key.
   Solution:
   Sports are split into
   a) Sports(sport_id,sport_name)
   b) SportType(sport_name , type).

```sql
-- ----------------------------------------------------------
-- 1. STRONG ENTITY SETS
-- ----------------------------------------------------------

CREATE TABLE Person (
   person_id VARCHAR(10) PRIMARY KEY,
   person_name VARCHAR(100) NOT NULL,
   gender CHAR(1), -- 'M', 'F', 'O
   dob DATE,
   contact_no VARCHAR(15),
   collage_name VARCHAR(100),
   roles VARCHAR(50)
);

CREATE TABLE Tournament (
   tournament_id VARCHAR(10) PRIMARY KEY,
   tournament_year INT NOT NULL,
   season VARCHAR(50),
   start_date DATE,
   end_date DATE
);

CREATE TABLE Sponsors (
   sponsor_id VARCHAR(10) PRIMARY KEY,
   name VARCHAR(100) NOT NULL,
   contact_no VARCHAR(15),
   address VARCHAR(255),
   company VARCHAR(100)
);

CREATE TABLE Sports (
   sport_id VARCHAR(10) PRIMARY KEY,
   sport_name VARCHAR(100) NOT NULL UNIQUE,
   type VARCHAR(50) -- e.g., 'Team', 'Individual'
);

CREATE TABLE Equipments (
   equipment_id VARCHAR(10) PRIMARY KEY,
   equipment_name VARCHAR(100) NOT NULL,
   number INT
);

CREATE TABLE Coach (
   coach_id VARCHAR(10) PRIMARY KEY,
```

```
    coach_name VARCHAR(100) NOT NULL
);

CREATE TABLE Venue (
    venue_id VARCHAR(10) PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    location VARCHAR(255)
);

CREATE TABLE Referee (
    referee_id VARCHAR(10) PRIMARY KEY,
    referee_name VARCHAR(100) NOT NULL,
    contact_no VARCHAR(15)
);

CREATE TABLE Organizer (
    member_id VARCHAR(10) PRIMARY KEY,
    member_name VARCHAR(100) NOT NULL,
    contact_no VARCHAR(15)
);

-- ------------------------------------------------------------
-- 2. SPECIALIZED ENTITY SETS (Subclasses of Person)
-- ------------------------------------------------------------

CREATE TABLE Player (
    player_id VARCHAR(10) PRIMARY KEY,
    height DECIMAL(5, 2),
    weight DECIMAL(5, 2),
    bloodgroup VARCHAR(5),
    joining_year INT,
    FOREIGN KEY (player_id) REFERENCES Person(person_id)
);

CREATE TABLE Spectator (
    spectator_id VARCHAR(10) PRIMARY KEY,
    membership_type VARCHAR(50),
    FOREIGN KEY (spectator_id) REFERENCES Person(person_id)
);

-- ------------------------------------------------------------
-- 3. CORE ENTITIES WITH FOREIGN KEYS
-- ------------------------------------------------------------
```

```sql
CREATE TABLE Team (
    team_id VARCHAR(10) PRIMARY KEY,
    sport_id VARCHAR(10) NOT NULL,
    team_name VARCHAR(100) NOT NULL,
    collage_id VARCHAR(100),
    caption_id VARCHAR(10),
    FOREIGN KEY (sport_id) REFERENCES Sports(sport_id),
    FOREIGN KEY (caption_id) REFERENCES Player(player_id)
);

-- Renamed from Match to Game
CREATE TABLE Game (
    match_id VARCHAR(10) PRIMARY KEY,
    sport_id VARCHAR(10) NOT NULL,
    tournament_id VARCHAR(10) NOT NULL,
    date DATE NOT NULL,
    match_type VARCHAR(50),
    time TIME,
    venue_id VARCHAR(10),
    referee_id VARCHAR(10),
    FOREIGN KEY (sport_id) REFERENCES Sports(sport_id),
    FOREIGN KEY (tournament_id) REFERENCES Tournament(tournament_id),
    FOREIGN KEY (venue_id) REFERENCES Venue(venue_id),
    FOREIGN KEY (referee_id) REFERENCES Referee(referee_id)
);

-- Renamed from Rules to SportRules
CREATE TABLE SportRules (
    sport_id VARCHAR(10) PRIMARY KEY,
    rules TEXT,
    FOREIGN KEY (sport_id) REFERENCES Sports(sport_id)
);

CREATE TABLE Accommodation (
    person_id VARCHAR(10) NOT NULL,
    tournament_id VARCHAR(10) NOT NULL,
    room_no VARCHAR(10),
    check_in_date DATE,
    check_out_date DATE,
    status VARCHAR(50),
    PRIMARY KEY (person_id, tournament_id),
    FOREIGN KEY (person_id) REFERENCES Person(person_id),
    FOREIGN KEY (tournament_id) REFERENCES Tournament(tournament_id)
);
```

```
-- ----------------------------------------------------------
-- 4. RELATIONSHIP SETS (Junction/Linking Tables)
-- ----------------------------------------------------------

CREATE TABLE OrganizeTournament (
   tournament_id VARCHAR(10) NOT NULL,
   member_id VARCHAR(10) NOT NULL,
   role VARCHAR(50),
   department VARCHAR(50),
   PRIMARY KEY (tournament_id, member_id),
   FOREIGN KEY (tournament_id) REFERENCES Tournament(tournament_id),
   FOREIGN KEY (member_id) REFERENCES Organizer(member_id)
);

CREATE TABLE SpectatorViewGame (
   spectator_id VARCHAR(10) NOT NULL,
   match_id VARCHAR(10) NOT NULL,
   PRIMARY KEY (spectator_id, match_id),
   FOREIGN KEY (spectator_id) REFERENCES Spectator(spectator_id),
   FOREIGN KEY (match_id) REFERENCES Game(match_id)
);

CREATE TABLE PlayerPlaysGame (
   player_id VARCHAR(10) NOT NULL,
   match_id VARCHAR(10) NOT NULL,
   PRIMARY KEY (player_id, match_id),
   FOREIGN KEY (player_id) REFERENCES Player(player_id),
   FOREIGN KEY (match_id) REFERENCES Game(match_id)
);

CREATE TABLE SportEquipments (
   sport_id VARCHAR(10) NOT NULL,
   equipment_id VARCHAR(10) NOT NULL,
   number INT,
   PRIMARY KEY (sport_id, equipment_id),
   FOREIGN KEY (sport_id) REFERENCES Sports(sport_id),
   FOREIGN KEY (equipment_id) REFERENCES Equipments(equipment_id)
);

CREATE TABLE PlayerSport (
   player_id VARCHAR(10) NOT NULL,
   sport_id VARCHAR(10) NOT NULL,
   level VARCHAR(50),
```

```
    experience_years INT,
    PRIMARY KEY (player_id, sport_id),
    FOREIGN KEY (player_id) REFERENCES Player(player_id),
    FOREIGN KEY (sport_id) REFERENCES Sports(sport_id)
);

CREATE TABLE PlayerTeam (
    player_id VARCHAR(10) NOT NULL,
    team_id VARCHAR(10) NOT NULL,
    joining_date DATE,
    end_date DATE,
    PRIMARY KEY (player_id, team_id),
    FOREIGN KEY (player_id) REFERENCES Player(player_id),
    FOREIGN KEY (team_id) REFERENCES Team(team_id)
);

CREATE TABLE TeamPlaysGame (
    match_id VARCHAR(10) NOT NULL,
    team_id VARCHAR(10) NOT NULL,
    PRIMARY KEY (match_id, team_id),
    FOREIGN KEY (match_id) REFERENCES Game(match_id),
    FOREIGN KEY (team_id) REFERENCES Team(team_id)
);

CREATE TABLE TeamCoach (
    team_id VARCHAR(10) NOT NULL,
    coach_id VARCHAR(10) NOT NULL,
    join_date DATE,
    end_date DATE,
    PRIMARY KEY (team_id, coach_id),
    FOREIGN KEY (team_id) REFERENCES Team(team_id),
    FOREIGN KEY (coach_id) REFERENCES Coach(coach_id)
);

CREATE TABLE PlayerStatistics (
    player_id VARCHAR(10) NOT NULL,
    match_id VARCHAR(10) NOT NULL,
    status_name VARCHAR(50) NOT NULL,
    status_value VARCHAR(50),
    PRIMARY KEY (player_id, match_id, status_name),
    FOREIGN KEY (player_id) REFERENCES Player(player_id),
    FOREIGN KEY (match_id) REFERENCES Game(match_id)
);
```

```sql
CREATE TABLE TeamStatistics (
    team_id VARCHAR(10) NOT NULL,
    match_id VARCHAR(10) NOT NULL,
    status_name VARCHAR(50) NOT NULL,
    status_value VARCHAR(50),
    PRIMARY KEY (team_id, match_id, status_name),
    FOREIGN KEY (team_id) REFERENCES Team(team_id),
    FOREIGN KEY (match_id) REFERENCES Game(match_id)
);

-- Renamed from Result to GameOutcome
CREATE TABLE GameOutcome (
    match_id VARCHAR(10) NOT NULL,
    team_id VARCHAR(10) NOT NULL,
    outcome VARCHAR(50), -- e.g., 'Win', 'Loss', 'Draw'
    PRIMARY KEY (match_id, team_id),
    FOREIGN KEY (match_id) REFERENCES Game(match_id),
    FOREIGN KEY (team_id) REFERENCES Team(team_id)
);
```